

ECS 189G Spring 2026: Multimodal Foundation Models

TO PREPARE AND SUBMIT HOMEWORK

Follow these steps exactly, so your assignment can be properly graded:

1. Carefully read the README.MD in the code.
2. Complete your Python scripts (e.g., `eval_classification.py` and `retrieval.py`).
3. Compile the overall accuracy of your zero-shot classification, your Prompt Engineering accuracy comparison and analysis, and the top-5 retrieved images from Task 5 with your qualitative analysis into a single PDF file named `Report.pdf`.
4. Package your Python scripts, and your `Report.pdf` into a single `.zip` archive.
5. Upload your `.zip` file to the course submission portal by the due date.

INSTRUCTIONS

In this assignment, you will explore the visual-language alignment capabilities of the pre-trained **CLIP (Contrastive Language-Image Pre-training)** model in a strictly training-free (zero-shot) setting. Your goal is to evaluate its performance on image classification and build a cross-modal text-to-image retrieval system from scratch.

- **Dataset:** You will use the Imagenette dataset for zero-shot classification and a miniature subset of MS-COCO (approx. 5,000 images) for cross-modal retrieval.
- **Task:** This assignment uses the pre-trained CLIP model (`openai/clip-vit-base-patch32`). Since no gradient updates or backpropagation are required, all tasks can be efficiently executed on standard CPUs by relying purely on forward passes and matrix operations (cosine similarity).

ENVIRONMENT SETUP

It is highly recommended to use a virtual environment (e.g., Conda or Python venv) to avoid dependency conflicts. We have provided a `requirements.txt` file for you.

Zero-Shot Image Classification [40 POINTS]

Create a new Python script (e.g., `eval_classification.py`) :

- **Extract image embeddings via the Vision Encoder and text embeddings via the Text Encoder.**
- **Compute the cosine similarities between a given image and all 10 possible class labels (using matrix multiplication) to predict the correct category.**
- **Evaluate and report the overall accuracy on the validation set of Imagenette.**

Visual Prompt Engineering [10 POINTS]

Evaluate how text prompt formatting impacts visual classification.

- **Implement and compare the zero-shot classification accuracy of at least two prompt template variants: bare labels (e.g., "dog"), a standard instructional template (e.g., "a photo of a dog"), and ensemble templates.**
- **Provide analysis in your report explaining why context matters for pre-trained vision-language models.**

Text-to-Image Retrieval System [50 POINTS]

Build a cross-modal search engine (e.g., `retrieval.py`):

- [20 points]: Iterate through the database of approximately 5,000 COCO images, pre-compute their image embeddings, and store them. (do not upload the downloaded images and their embeddings when you submit your homework)
- [20 points]: Implement a top-k retrieval function that takes a complex, open-ended text query (e.g., "a dog catching a frisbee in the air"), computes similarities against the database, and visualizes the top-5 matching images.
- [10 points]: Visualization & Analysis: Use a library like matplotlib to plot the top 5 images for at least 3 distinct, creative text queries. Include a qualitative analysis of some cases in your report, discussing instances where the model retrieves incorrect but conceptually similar images.